Kantorovich's Problem:

(Discrete version)

$$\min_{\pi \in \Pi(\mathbf{a},\mathbf{b})} \sum_{i=1}^{n} \sum_{j=1}^{m} \mathbf{C}_{i,j}\, \pi_{i,j}$$

Monge's Problem:

(Matching/permutation version)

$$\min_{\sigma \in \Sigma_n} \frac{1}{n} \sum_{i=1}^{n} \mathbf{C}_{i,\sigma(i)}$$

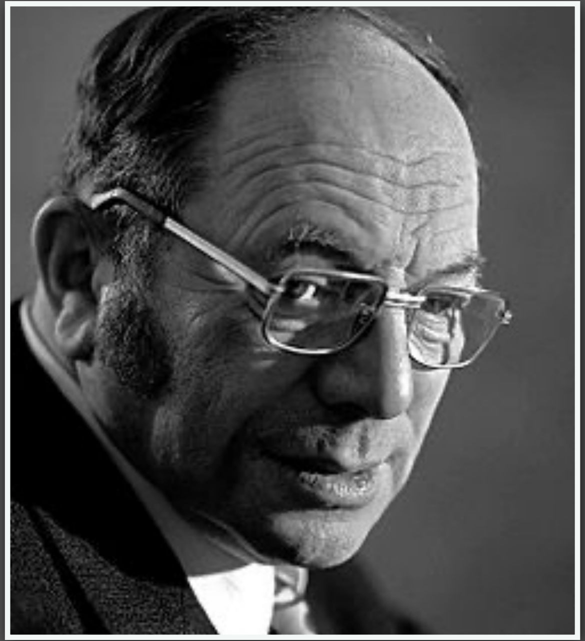
Pop-Quiz

Consider the uniform case $\mathbf{a} = \frac{1}{n}\mathbf{1}_n, \mathbf{b} = \frac{1}{n}\mathbf{1}_n$.

Which one is larger? $\Pi(\mathbf{a}, \mathbf{b})$ or Perm(n)?

$$\text{Perm}(n) \subset \Pi(\mathbf{a}, \mathbf{b})$$

Note: this implies that

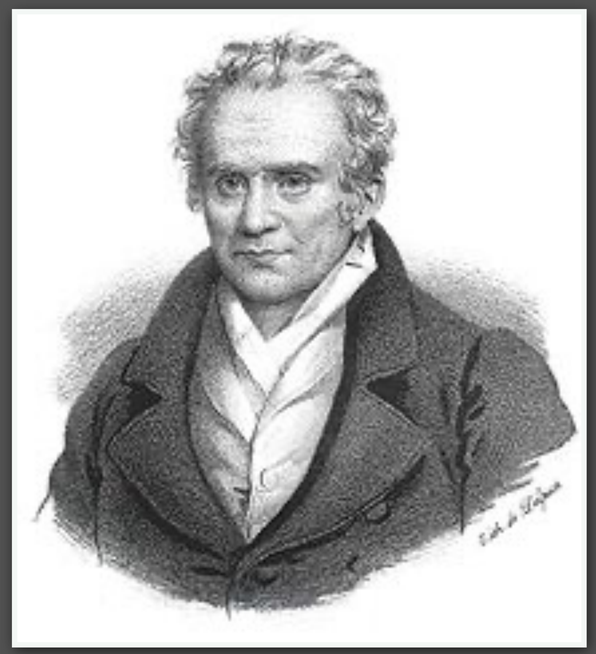


Kantorovich's Problem:

(Discrete version)

$$\min_{\pi \in \Pi(\mathbf{a},\mathbf{b})} \sum_{i=1}^{n} \sum_{j=1}^{m} \mathbf{C}_{i,j}\ \pi_{i,j}$$

$$\leq$$



Monge's Problem:

(Matching/permutation version)

$$\min_{\sigma \in \Sigma_n} \frac{1}{n} \sum_{i=1}^{n} \mathbf{C}_{i,\sigma(i)}$$

Birkhoff-von Neuman Theorem